

Portfolio → “360 Real Estate Compass” — Capability Map

Prepared for: Prof. Ingeborg Rocker, I2CE Lab (GT College of Design) **From:** Sam Duong — urban-AI / digital-twin stack (GT) **One line:** We already run working prototypes of the Compass’s *core engine* and three of its hardest data layers. The funded work is **integration + validation on a real Atlanta site (The Stitch)** — not building from zero.

How the existing repos map onto the Compass

360 Compass layer	Existing asset (repo)	Maturity	What the funded team adds
Estimate value / optimal location (the core)	polyscape — XGBoost + SHAP/GeoShapley site-suitability prediction, multi-scale viz	Working prototype	Retarget the model from a composite <i>suitability</i> index to assessed/transaction value (add real sales data)
GIS / geospatial mapping	polyscape, nihonbashi twin, polymetron	Working	Unify on one basemap/coordinate frame
Demographic & economic trends	polyscape (Census ACS + LODS jobs)	Working	Add rental / market-demand feeds
Urban growth / neighborhood evolution	polyscape (predictive hex surfaces)	Working	Time-series / trend layer
Infrastructure & mobility access	nihonbashi networks, polyscape (OSMnx/Overture), heatwave_route	Working	Transit/access scoring
Environmental & climate factors	nihonbashi-robot-sim — heat digital twin + scenario A/B + living visualization	Working (synthetic + OSM geometry)	Plug in real Atlanta heat / land-surface-temp data
Zoning, land-use & building code	smart-codes-cura — multi-agent RAG + knowledge graph	Demo	Atlanta zoning/entitlement focus

360 Compass layer	Existing asset (repo)	Maturity	What the funded team adds
	over codes across 9 US cities (GT CURA)		
Existing-conditions / site survey	polymetron / polymetron-ar — on-device CV + VLM scene assessment	Working demo	Structured condition reports
Generative AI → development variations / adaptive reuse	agentic + optimization scaffolding (cuOpt, MCP) + code-RAG + perception	Foundation	The generative dev-scenario engine
Construction / development costs	—	gap	Source cost data / partner
Comparable sales / investment performance	polyscape (comparative framework)	Partial	Real comps feed

Coverage: working prototypes touch ~8 of 11 Compass layers; genuine gaps are construction cost, deep market comps, and a **unified integration layer**.

The three pillars to demo

1. **polyscape** — *the Compass engine*. Predicts where to develop and *explains why* (SHAP), at city → district → street zoom. This is Rocker’s “estimate value / identify optimal locations” goal, already running.
2. **nihonbashi-robot-sim** — *the climate + living-twin module*. Interactive, time-of-day, heat-aware district model — beats a static render or wood model for usability questions.
3. **smart-codes-cura** — *the code/zoning brain*. Answers cross-jurisdiction zoning & building-code questions via RAG. Already GT CURA-branded → inside the GT ecosystem she’s recruiting into.

Supporting: **polymetron** (perception / existing-conditions), **heatwave_route** (mobility + climate), **Food_Circular_Network** (circular-economy thread — aligns with I2CE’s literal name).

Proposed pilot: The Stitch (division of labor)

Our stack delivers	I2CE / architecture studio delivers
Location-intelligence + value/suitability scoring (polyscape)	Urban-design, massing, density/height scenarios

Our stack delivers	I2CE / architecture studio delivers
Climate/usability analysis of the new deck-park public realm (nihonbashi twin)	Vision report, entitlement framework
Zoning/code screening (smart-codes)	Physical wood massing model, exhibition materials
Interactive district-scale digital model for stakeholder & fundraising	Comprehensive redevelopment vision

The Stitch is an ideal pilot for us: it caps a highway with new public space, so the key question — *“will people use this ground in an Atlanta summer?”* — is exactly our climate-twin strength.

What the I2CE sponsorship funds

These are **separate prototypes today**, not one platform. A small funded team across summer/fall would **integrate them into a single “Compass” pilot and validate it on The Stitch** — matching Rocker’s timeline (robust pilot by end of summer, real-site test in fall).

Maturity & honesty ledger (don’t overstate in the room)

Asset	Honest status
polyscape	Working prototype. Predicts a composite suitability index (POI mix, walk score, etc.), not validated property value yet — retargetable with real sales data.
smart-codes-cura	RAG prototype for cross-jurisdiction code Q&A; not a certified compliance tool.
nihonbashi twin	Infrastructure complete; runs on synthetic + OSM data. Photoreal 3D mesh is Tokyo/PLATEAU-only — Atlanta would use OSM/USGS. Research <i>result</i> pending real heat data.
polymetron / -ar	Working on-device CV/VLM demo — scene description/flagging, not a calibrated inspection instrument.
heatwave_route	Working heat-routing tool (Tokyo studio).
Integration layer	Does not exist yet — this is the proposed funded deliverable.

